

D784

Continuous Integration and Continuous Delivery

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A deployment declares 6 replicas of a checkout service. The cluster runs on nodes bought last year, and the service is written in Go. One node fails and takes 2 replicas with it. With no one stepping in, what does the orchestrator do?

- A. Changes the desired count to 4 to match what is running.
- B. Starts 6 new replicas, for 10 in total.
- C. Starts 2 new replicas on healthy nodes to return to 6.
- D. Runs with 4 until an admin redeploys.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A new container takes 90 seconds to load a large cache before it can answer requests. Users routed to it during that time get errors. The team's dashboards use a popular open-source tool. Which change fixes the errors?

- A. Double the number of replicas.
- B. Add a liveness probe that fails until the cache loads.
- C. Switch to a different dashboard tool.
- D. Add a readiness probe so traffic arrives only after the cache is loaded.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A container's process is still running but has deadlocked and will never recover. Which probe makes the orchestrator restart it?

- A. No probe; the orchestrator always detects every unhealthy process.
- B. A liveness probe.
- C. A readiness probe.
- D. A startup probe alone.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A build server's jobs were set up years ago by clicking through its web interface. The admin who set them up has left, the server sits in the main data center, and no one knows why one branch builds differently. What is the best fix?

- A. Move the build server to the cloud.
- B. Lock the server so no one can ever change a setting again.
- C. Take screenshots of every UI setting and save them in a document.
- D. Define the build steps in a pipeline script stored in the repository with the code.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A pipeline runs unit tests, integration tests against a test database, and then a code-coverage check with an 80% floor. A build passes all unit and integration tests but shows 72% coverage. The commit came from the team lead. What should the pipeline do?

- A. Lower the floor to 72% so the build passes.
- B. Promote it, because all tests passed.
- C. Stop the build and not promote it.
- D. Promote it, because the team lead wrote it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A team rebuilds its application from source separately for test, staging, and production. A bug shows up only in production even though the same commit passed staging. Staging deploys run on a nightly schedule. What change best prevents this?

- A. Build one versioned artifact once and promote that same artifact through every environment.
- B. Deploy to staging every hour instead of nightly.
- C. Keep separate builds so each is tuned for its environment.
- D. Add more tests to the production build only.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An online store runs version 1 on its blue environment. It deploys version 2 to an identical green environment, tests it, and switches the load balancer to green. Ten minutes later error rates spike. The release was scheduled for a quiet Tuesday morning. What is the fastest way to restore service?

- A. Restore blue from last night's backups, then switch.
- B. Switch the load balancer back to blue.
- C. Wait, since quiet Tuesday traffic means few users are hurt.
- D. Build a hotfix for version 2 and redeploy green.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Compared with updating servers in place, what is the main cost that blue-green deployment adds?

- A. Downtime is always required during the switch.
- B. Slower rollback than in-place updates.
- C. Sending a small share of users to the new version first.
- D. Running two full production environments, roughly double the capacity during a release.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A streaming service with 400,000 active users releases a new version to 5% of traffic first. The release was built by a team in another time zone. How many users see the new version in the first stage, and what should decide whether it expands?

- A. 20,000; its error and latency metrics compared with the current version.
- B. 20,000; the planned calendar date for full rollout.
- C. 20,000; sign-off from the team in the other time zone.
- D. 380,000; its error and latency metrics compared with the current version.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A canary stage sent a new version to 3,600 users out of 120,000. What percentage of users was in the canary?

- A. 97%
- B. 3%
- C. 0.03%
- D. 33.3%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A team measures lead time for changes from commit to running in production. In one sprint, five changes took 2, 4, 6, 8, and 30 hours. The sprint had 3 planning meetings. What is the median lead time, and why report it?

- A. 30 hours; the slowest change sets lead time.
- B. 3 hours; one per planning meeting.
- C. 6 hours; the median is not pulled up by the one 30-hour outlier.
- D. 10 hours; the average of all five changes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

After a team added a weekly manual approval board, lead time for changes rose from 1 day to 5 days and deployment frequency fell. What does this suggest?

- A. The weekly board is a bottleneck that batches changes and slows delivery.
- B. Nothing; process changes can never affect lead time.
- C. Delivery sped up, because a larger lead time number is better.
- D. Stability improved, so the change was a success.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A deployment declares 6 replicas of a checkout service. ~~The cluster runs on nodes bought last year, and the service is written in Go.~~ One node fails and takes 2 replicas with it. With no one stepping in, what does the orchestrator do?

Target word: orchestrator response to lost replicas **Decides:** Declared state is 6 replicas.

A	Inverted operation	Reconciliation changes reality to match the declaration, not the reverse.
B	Wrong stem number	Uses the declared 6 as the number to add instead of the gap of 2.
C	Correct	The controller reconciles actual state back to declared state.
D	Dropped element	Drops the self-healing loop that orchestration provides.

2. Answer D Concept 2

A new container takes 90 seconds to load a large cache before it can answer requests. Users routed to it during that time get errors. ~~The team's dashboards use a popular open source tool.~~ Which change fixes the errors?

Target word: stop traffic to a warming container **Decides:** The container is alive but not yet ready.

A	Adjacent category	More replicas still receive traffic during warm-up.
B	Confusion set	Liveness failures trigger restarts, which can loop forever during warm-up.
C	Camouflage	Dashboards observe errors; they do not route traffic.
D	Correct	Readiness controls whether the container is in the traffic pool.

3. Answer B Concept 2 · transfer

A container's process is still running but has deadlocked and will never recover. Which probe makes the orchestrator restart it?

Target word: which probe restarts a hung container **Decides:** The process needs a restart, not just traffic removal.

A	Absolute language	Without a probe it sees only process exit, and a deadlocked process has not exited.
B	Correct	A failing liveness probe tells the orchestrator to restart the container.
C	Confusion set	It only removes the container from traffic; it stays deadlocked.
D	Adjacent category	Startup probes guard the first launch, not a hang that comes later.

4. Answer D Concept 3

A build server's jobs were set up years ago by clicking through its web interface. The admin who set them up has left, ~~the server sits in the main data center,~~ and no one knows why one branch builds differently. What is the best fix?

Target word: make builds repeatable and understood **Decides:** Build logic lives only in hand-clicked settings.

A	Camouflage	Location does not make the configuration repeatable.
B	Absolute language	Freezes the mystery in place and blocks needed changes.
C	Dropped element	Documents the settings but keeps them manual and unversioned.
D	Correct	Pipeline as code is versioned, reviewed, and rebuildable.

5. Answer C Concept 4

A pipeline runs unit tests, integration tests against a test database, and then a code-coverage check with an 80% floor. A build passes all unit and integration tests but shows 72% coverage. ~~The commit came from the team lead.~~ What should the pipeline do?

Target word: promotion decision **Decides:** A gate (coverage floor) failed.

A	Inverted operation	Moves the gate to fit the build instead of fixing the build.
B	Dropped element	Ignores the coverage gate.
C	Correct	Every gate must pass; one failure stops promotion.
D	Camouflage	Author seniority does not override a gate.

6. Answer A Concept 5

A team rebuilds its application from source separately for test, staging, and production. A bug shows up only in production even though the same commit passed staging. ~~Staging deploys run on a nightly schedule.~~ What change best prevents this?

Target word: prevent environment-only bugs from rebuilds **Decides:** Each environment ran a different build.

A	Correct	What was tested is exactly what ships; only configuration changes.
B	Camouflage	Frequency does not make the builds identical.
C	Inverted operation	That is the cause of the difference.
D	Adjacent category	Tests a build that still differs from the one staged.

7. Answer B Concept 6

An online store runs version 1 on its blue environment. It deploys version 2 to an identical green environment, tests it, and switches the load balancer to green. Ten minutes later error rates spike. ~~The release was scheduled for a quiet Tuesday morning.~~ What is the fastest way to restore service?

Target word: fastest rollback **Decides:** Blue is still running version 1.

A	Dropped element	Forgets that blue is still intact and running.
B	Correct	Blue-green rollback is a traffic switch to the untouched old environment.
C	Camouflage	Low traffic does not justify leaving errors when rollback is instant.
D	Adjacent category	Rolling forward is slower than switching back.

8. Answer D Concept 6 • transfer

Compared with updating servers in place, what is the main cost that blue-green deployment adds?

Target word: cost of blue-green **Decides:** Two full environments must exist.

A	Absolute language	The switch is designed to avoid downtime.
B	Inverted operation	Rollback is faster: a traffic switch.
C	Confusion set	That describes a canary release.
D	Correct	The idle environment is what makes instant rollback possible.

9. Answer A Concept 7

A streaming service with 400,000 active users releases a new version to 5% of traffic first. ~~The release was built by a team in another time zone.~~ How many users see the new version in the first stage, and what should decide whether it expands?

Target word: canary size and expansion rule **Decides:** 5% of 400,000; expand on health metrics.

A	Correct	$0.05 \times 400,000 = 20,000$; canaries expand only if healthy.
B	Dropped element	Drops the metric comparison that is the whole point of a canary.
C	Camouflage	Where the builders sit does not decide canary health.
D	Inverted operation	That is the share still on the old version.

10. Answer B Concept 7 · transfer

A canary stage sent a new version to 3,600 users out of 120,000. What percentage of users was in the canary?

Target word: solve for the canary percentage **Decides:** $3,600 / 120,000$.

A	Confusion set	The share not in the canary.
B	Correct	$3,600 / 120,000 = 0.03 = 3\%$.
C	Dropped element	Drops the $\times 100$ when converting to a percent.
D	Inverted operation	Divides 120,000 by 3,600.

11. Answer C Concept 8

A team measures lead time for changes from commit to running in production. In one sprint, five changes took 2, 4, 6, 8, and 30 hours. ~~The sprint had 3 planning meetings.~~ What is the median lead time, and why report it?

Target word: median lead time and its purpose **Decides:** One 30-hour outlier skews the mean.

A	Wrong stem number	The maximum, not the median.
B	Camouflage	Meetings do not decide lead time.
C	Correct	Sorted values 2, 4, 6, 8, 30 have middle value 6.
D	Adjacent category	That is the mean, which the outlier inflates.

12. Answer A Concept 8 · transfer

After a team added a weekly manual approval board, lead time for changes rose from 1 day to 5 days and deployment frequency fell. What does this suggest?

Target word: read the metric change **Decides:** Higher lead time and fewer deploys mean slower flow.

A	Correct	Both throughput metrics got worse after the gate was added.
B	Absolute language	Approval waits are part of lead time.
C	Inverted operation	Lower lead time is better.
D	Confusion set	Lead time and deploy frequency measure throughput, not stability.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Desired state and self-healing	An orchestrator keeps the running state equal to the declared state, replacing lost replicas without human action.	Thinking the orchestrator waits for an admin, or lowers the target to match reality.	A declared replica count and a node failure.
2	Readiness vs. liveness probes	A readiness probe decides whether a container gets traffic; a liveness probe decides whether it gets restarted.	Using a liveness probe to handle slow startup, which causes restart loops.	Slow warm-up (readiness) vs. a hung process (liveness).
3	Pipeline as code	Build and deploy steps live in a script in version control beside the application, so they are repeatable, reviewable, and recoverable.	Documenting click-configured build jobs instead of scripting them.	Build jobs set up by hand in a UI that nobody understands.
4	Quality gates	A failed gate (tests, coverage, scans) stops promotion; all gates must pass, not just some.	Promoting because the main tests passed, or lowering the bar to pass.	A build that passes some checks and misses a threshold.
5	Build once, promote the same artifact	Build a single versioned artifact once and move that same artifact through every environment, changing only configuration.	Rebuilding from source for each environment.	A bug that appears only in production for the same commit.
6	Blue-green deployment	Run two identical environments and switch traffic between them; rollback is switching back, at the cost of double capacity.	Thinking rollback requires a rebuild or restore.	Two named environments and a load balancer switch.
7	Canary release	Send a small share of traffic to the new version, compare its metrics with the current version, and expand only if healthy.	Expanding on a schedule regardless of metrics.	A percentage of users getting the new version first.
8	Lead time for changes	Lead time runs from commit to production; use the median for skewed data, and falling lead time means faster flow.	Treating longer lead time as better, or using the mean on skewed data.	A list of commit-to-production durations with one outlier.

Course scope (public description): In this course, students will learn advanced DevOps practices essential for modern software development environments. Students will master container orchestration techniques to enhance application scalability, resilience, and high availability. They will learn to configure build servers using automated scripts, integrating robust integration testing and quality checks seamlessly into the development pipeline. Students will develop comprehensive release management plans designed to ensure smooth and reliable software releases with minimal downtime. They will explore the implementation of automated deployment pipelines, enabling continuous delivery and facilitating rapid, incremental releases of software updates. Students will gain insights into analyzing performance indicators to optimize software development and delivery processes continuously.